

Terminal Accuracy Without Confident Convergence: A Finite-Budget Simulation Audit of Model-Defined Chemistry Diagnosis

Structured Abstract

Background: Short diagnostic budgets can yield a correct model-defined terminal ranking without satisfying the system's confidence-based stopping rule.

Objective: We audited prerequisite-probe selection, fixed insertion, one pre-specified generator perturbation, and the terminal-versus-convergence gap. Secondary analyses examined Persona-v2 answer-state shifts and robustness and supply-constrained prescription mapping.

Methods: The implementation-bound simulation crossed 27 chemistry targets, four states, three arms, 50 paired replicates, and two generators (32,400 intended journeys). Arms shared noise streams. Ten thousand target-stratified paired bootstraps quantified Monte Carlo uncertainty conditional on this fixed catalog, not population uncertainty. Secondary analyses used 50 persona clusters and a two-target prescription model.

Results:

At 15 items, adaptive P-state correct convergence was 148/1,150 (12.9%) versus 9/1,150 (0.8%) for the local ladder; the paired difference was 12.1 points (95% CI 10.3% to 13.8%). H1 was partially supported because the adaptive rate missed the frozen 50% criterion. At nine items, fixed insertion minus adaptive C-state misdiagnosis was 5.7 points, while adaptive minus local was 9.7 points; H2 was not supported. H3 was supported, and the direction-only H4 check passed.

Persona v2 (50 clusters; 3/6 providers eligible): conditional-accuracy shift=13.4% (95% CI 9.8% to 17.4%); blind agreement=48.0%-79.0%, answer-category/canonical-complete-output stability=95.0%/11.7%, failure=0.0%; Automated error coding was unavailable: GPT judge failed and Claude judge was unavailable; pairwise judge agreement was not estimable. P2 (retrospectively byte-anchored illustration; historical order unproven; 2 targets/8 rows/3 sources): mismatched minutes A/B/C=3.298/4.030/1.525; Arm-C structural-failure fraction=0.500.

Conclusions: Prerequisite probes improved one contrast without reliable confident convergence, and adaptive selection failed no-harm. In this implementation, terminal accuracy overstated operational convergence. The study does not estimate learning gains, efficacy, or human behavioral validity.

Keywords

adaptive diagnosis; computerized adaptive testing; chemistry education; Bayesian learner model; simulation audit; stopping rule; prerequisite probe; model-defined remediation

1. Introduction

Adaptive educational systems often compress a sequence of responses into a mastery score and then use that score to choose the next item or learning resource. That loop is useful when the only decision is whether to continue practice. It is less adequate when the downstream action depends on *which model-defined condition* produced the same observed error. A low score may route a learner to a prerequisite explanation, a worked reasoning bridge, or a direct introduction to the target concept. If the diagnostic distinction is weak, a confident-looking posterior can spend a limited learning budget on the wrong type of resource.

This paper studies that measurement problem inside a local Shanghai high-school chemistry diagnostic and recommendation implementation. The system represents a target concept using four model-defined diagnostic states: mastered (M), prerequisite gap (P), reasoning-chain instability (C), and unlearned target knowledge (U). The labels are operational hypotheses in an engineered model. They are not clinical categories, psychometric traits established in people, or observations of human participants. Their value depends on whether the response channel can distinguish them within the item budget used by the product and whether a state-conditioned downstream action is better aligned with the model-defined resource role than a state-agnostic one.

The motivating tension is straightforward. For local target questions, the production likelihoods assigned to P and U differ by only 0.10. With arbitrarily many independent observations under the model, that gap can identify the states. With 9, 15, or 25 questions, the posterior may rank P first without crossing the production confidence and gap thresholds. A prerequisite question creates a different response channel that separates P from U more strongly, but the same channel can blur C with other states. The design question is therefore not whether prerequisite questions are informative in general. It is when a selector should expose that channel and whether the stopping rule recognizes the evidence it accumulates.

The study was developed in response to known instrument limitations, including multiple-choice-heavy evidence, sparse multi-step and media-dependent coverage, and uncertain mapping from observed error to downstream remediation. These limitations make a large efficacy claim premature. They also create a legitimate audit question: under a fixed model, catalog, and response budget, where does state recovery fail and how does that failure constrain later decisions? We call this

a **budget-limited confident-convergence failure**. The phrase is deliberately finite and operational. It does not assert structural or asymptotic impossibility.

The primary programmatic experiment addresses four frozen hypotheses. H1 asks whether belief-triggered expected information gain (EIG) improves correct confident convergence for P relative to a local-only ladder while also reaching a minimum absolute rate. H2 asks whether a fixed prerequisite quota harms C relative to the adaptive policy and whether the adaptive policy stays within a no-harm margin relative to the local-only ladder. H3 is a subordinate adaptive-selection sanity check. H4 tests whether the predicted H1 and H2 directions persist under one frozen family of generator misspecification. The decisions, supports, exclusions, seeds, and interval procedures were fixed before the confirmatory run.

Two later components are separated from that primary evidence. Persona v2 estimates paired deficit-control shifts in answer-state composition under controlled prompts and tests blind response robustness across language-model providers. Target-policy compliance is available only as a sparse six-row descriptive check. Persona v2 is an independent response-channel stress test, not an enlargement of the programmatic sample. P2 illustrates how diagnostic posteriors map to model-defined mechanically mismatched selected minutes and missed available-supply minutes within a small trusted resource library. Neither component is allowed to alter H1-H4.

The research questions are:

1. Does belief-triggered prerequisite probing improve P-state correct confident convergence over a local-only fixed ladder at the frozen item budget?
2. Does fixed-quota prerequisite probing raise C-state misdiagnosis relative to the adaptive arm, while the adaptive arm remains within the frozen no-harm margin relative to the local-only arm?
3. Do adaptive terminal accuracy and convergence-time advantages appear on the full target set, and do the predicted directions persist under the declared generator perturbation?
4. On the same target support, how large is the gap between a correct terminal argmax and a correct confidence-based stop, and what selector behavior is associated with that gap?
5. Secondarily and exploratorily, how do paired deficit-control answer states shift under controlled prompts, and how robust are blind terminal responses across providers and repeats?
6. Secondarily and illustratively, how do fixed diagnostic posteriors map to mechanically mismatched selected minutes, missed available-supply minutes, and structural failure under constrained trusted-video supply?

The contributions are correspondingly bounded. First, we provide a repository-frozen, pre-observation comparison of a belief-triggered selector, a local-only ladder, and a fixed prerequisite quota using the local product implementation. Second, we preserve paired random streams and co-report with-replacement stress and no-repeat common-support estimates rather than choosing the more favorable view. Third, we separate terminal classification from confident convergence and show that they can lead to very different assessments of readiness. Fourth, we expose the dependence of the findings on one declared response generator and on a narrow chemistry catalog. Finally, we pre-specify bounded secondary analyses of answer-state robustness and supply-constrained prescription mapping without allowing either to revise the primary decisions.

The claim boundary is central: this is a simulation-only audit of model-defined diagnostic states. It does not estimate learning gains, does not establish educational efficacy, does not establish human behavioral validity, and does not claim that the tested target set represents an entire chemistry curriculum.

2. Related Work

2.1 Adaptive testing and simulation

Computerized adaptive testing has a long history of using current evidence to select more informative items. Robbins-Monro formulations, early work on measurement efficiency, and educational applications established the basic logic well before the present system (Frederic M. Lord 1971; David J. Weiss 1982; David J. Weiss and G. Gage Kingsbury 1984). Methodological work shows why simulations are useful for comparing item-selection rules when the latent condition and response process are controlled (Juan Ramón Barrada et al. 2010; Kyung (Chris) Tyek Han 2018). Selection and termination are distinct design choices, and stopping rules can change test-length and precision tradeoffs (Seung W. Choi et al. 2011). The present study does not offer EIG as a new algorithm. It audits whether an existing selector aligns with a separately implemented confidence-based stopping rule.

Simulation can make failure modes observable because the generating state is known. It can also create circular evidence when the same likelihood family drives both the generator and inference. We address, but cannot eliminate, that risk by separating a matched generator from a pre-specified misspecified generator and by reporting the change between them. This design tests sensitivity to one perturbation family; it does not approximate the full range of response behavior found outside the model.

2.2 Learner models, confidence, and longitudinal scope

Bayesian knowledge tracing and related learner models estimate changing knowledge from observed practice (Albert T. Corbett and John R. Anderson 1994; Radek Pelánek 2017). Our primary experiment uses Bayesian updating but freezes the generating diagnostic state within each journey. It therefore evaluates state recovery during one diagnostic session, not knowledge acquisition, forgetting, transfer, or retention. The distinction matters: a model may update a posterior coherently without its state labels corresponding to stable constructs outside the simulator.

Spaced-practice scheduling concerns a different decision horizon (Behzad Tabibian et al. 2019; Junyao Ye et al. 2022). The product’s FSRS-inspired projection draws on documented software (Open Spaced Repetition contributors, n.d.; Anki contributors, n.d.), but neither scheduling research nor software documentation validates the likelihoods or stopping thresholds audited here.

2.3 Language-model response simulation

Recent studies use language models to instantiate diverse response profiles, review question items, or test tutoring agents (Xinyi Lu and Xu Wang 2024; Zhengyuan Liu et al. 2024; Hyoungwook Jin et al. 2025; Tao Wu et al. 2025; Alexander Scarlatos et al. 2025). This work demonstrates a potentially useful evaluation instrument, but it also makes construct boundaries essential. Surface plausibility and prompt compliance do not show that a response process matches human behavior. Direct tests of simulated tutoring dialogue have reinforced the distinction between apparent student-like language and substantive behavioral validity (Alexander Scarlatos et al. 2026).

Persona v2 therefore has two conditions with different purposes. Controlled prompts estimate paired deficit-control shifts in answer-state composition; the collapsed target map prevents general target-policy compliance from being a confirmatory outcome. Blind response robustness removes target labels and tests cross-provider consistency, technical failure, abstention, and repeat stability. Neither condition assigns independent sample status to provider calls or produces an authenticity score. Cross-model adjudication is automated descriptive coding, not a human reference standard.

2.4 From diagnostic output to remediation

A diagnostic posterior becomes educationally consequential only when it changes an action. In the current system, P, C, and U would route to different resource roles. This makes terminal state accuracy incomplete: policies may differ in stopping, abstention, or allocation of limited minutes. Budgeted coverage supplies a relevant algorithmic reference point (Samir Khuller et al. 1999), but P2 makes no approximation claim. It maps posterior-conditioned demand onto a fixed trusted library and reports model-defined mechanically mismatched selected and missed available-supply minutes. The illustration is a resource audit, not an optimizer benchmark or efficacy study.

3. Methods

3.1 System boundary and trusted catalog

The audited system serves Shanghai high-school chemistry. Its knowledge graph has 135 nodes, but the deterministic confirmatory catalog opened only 27 targets that passed service, scoring, mapping, media, and content-family gates. A mechanical prerequisite census admitted 23 of those targets to H1 and H2. Four targets lacked the required prerequisite-family structure and remained in the full descriptive and H3 grid but not in H1 or H2.

Questions were separated at both item and content-family levels. Two local families per target were reserved for held-out scoring and removed from all administered pools. Held-out responses were never used for posterior updates. The programmatic runner imported the production mastery updater and selector, so Bayesian updates, EIG scores, eligibility rules, and stopping behavior were the same implementations used by the local product. The experiment did not alter the official question bank or the frozen H1-H4 decisions.

3.2 Model-defined diagnostic states and response channels

For a target-local item of difficulty d , the production model assigns the state-wise probability of a correct response as

$$P(Y = 1 \mid M, P, C, U) = (0.90, \gamma + 0.10, 0.70 - 0.20d, \gamma),$$

where $\gamma = 0.25$ for multiple-choice items and 0.03 for numeric items. The local P and U channels therefore differ by 0.10. A prerequisite item uses

$$P(Y_{pre} = 1 \mid M, P, C, U) = (0.90, \gamma, 0.80, 0.75).$$

This second vector creates more separation between P and U, while making a correct prerequisite response compatible with several other states. The posterior starts uniform within each journey and updates after each observation. A terminal decision is the posterior argmax in the frozen M, P, C, U order. Correct confident convergence requires both the correct argmax and the production confidence and posterior-gap conditions. Exhausting the item budget is distinct from satisfying those conditions.

The prerequisite likelihood vector updates the target-state posterior directly. We therefore call the comparison a *probe-channel comparison*. It is not evidence that information propagated through the knowledge graph.

3.3 Programmatic design and item-selection arms

The intention-to-simulate grid crossed 27 targets, four fixed truth states, three arms, 50 replicates, and two response-generator conditions, yielding 32,400 planned journeys. A maximum 25-item run supplied frozen views at budgets 9, 15, and 25. When confidence was reached before a nominal view, the posterior and convergence time were carried forward without increasing the administered count.

The three arms were:

- **Arm A, belief-triggered EIG.** At each decision, all eligible local and prerequisite candidates competed under the production EIG selector. The production stopping rule was evaluated after every posterior update.
- **Arm B, local-only ladder.** Requested difficulties cycled through 0.25, 0.50, 0.75, and 1.00, with only local items eligible.
- **Arm C, fixed prerequisite quota.** Arm C used the same ladder but replaced every third position with a prerequisite item through position 24.

Fixed-arm ties were resolved deterministically by absolute difficulty distance, family identifier, and item identifier. After all unseen eligible families were exhausted, the frozen family-epoch replenishment rule permitted reuse. Repetition was measured and disclosed rather than treated as independent evidence.

The confirmatory hypotheses and decision branches were frozen before collection. H1 required both an Arm-A correct-convergence rate of at least 0.50 at budget 15 and a strictly positive lower confidence bound for A minus B. H2 required a strictly positive lower bound for C minus A C-state misdiagnosis at budget 9 and an upper bound below the +0.05 no-harm margin for A minus B. H3 required the adaptive arm not to lose terminal accuracy and not to take longer to converge than B. H4 required the H1 rescue and H2 fixed-quota harm point directions to remain positive under the frozen misspecified generator.

3.4 Response generators, pairing, and sparse-pool estimands

The matched generator sampled from the production probabilities. The misspecified generator instead drew item-level slip from a uniform 0.05 to 0.20 distribution, guess from a uniform 0.15 to 0.35 distribution, and one replicate-level ability offset from a zero-centered normal distribution with standard deviation 0.05, clipped to the interval -0.10 to 0.10. Inference retained its production constants in both conditions. Generator parameters were not passed to the posterior updater.

Arms were paired within target, truth state, generator condition, and replicate. They shared response-noise streams. The two held-out family outcomes were generated from seeds without an arm component, ensuring identical held-out draws for A, B, and C. This pairing reduced irrelevant Monte Carlo variation and permitted direct arm contrasts.

H1 and H2 used two pre-declared population views. The primary empirical stress view retained all 23 eligible targets and allowed the frozen family-replenishment behavior. The sensitivity view restricted analysis to the budget-specific three-arm common support on which every arm could complete without family repetition. That support contained nine targets at budget 9, four at budget 15, and one at budget 25. Both views were required regardless of direction.

Arm C could not construct the fixed prerequisite schedule for four targets. This produced 1,600 structural failures in the intended grid. Those failures were retained in the lifecycle and excluded uniformly from estimands under the frozen rule; they were not reclassified after results were known.

3.5 Outcomes, support identity, and statistical analysis

At each nominal budget, outcomes included correct confident convergence, converged misdiagnosis, terminal argmax accuracy, the four-by-four terminal and decision confusion matrices, convergence time, held-out Brier score, administered count, exact item repeats, and family repeats. Held-out Brier under the matched generator is an internal calibration diagnostic only.

Programmatic intervals used exactly 10,000 target-stratified paired-replicate percentile bootstrap resamples. Replicate identifiers were resampled with replacement within each target, while all arm observations for a sampled replicate remained paired. Contrasts were computed inside each resample. Reported quantities include the numerator, denominator, target count, weighting rule, and 95% interval.

The target catalog was fixed and was not resampled. These intervals therefore describe Monte Carlo uncertainty across simulator replicate streams conditional on the audited catalog. They are not uncertainty intervals for a population of chemistry targets, learners, schools, or diagnostic systems.

Support identity was checked before interpreting any pair. The 27-target full-set terminal and convergence rates are compared only with each other. The 23-target eligible-set rates are likewise paired only with the same eligible support. The original artifact registry encodes support through metric names, target counts, denominators, and predicates rather than a dedicated target-set hash. The final binder reconstructs and binds a dedicated target-set hash before insertion; portability still depends on preserving the frozen target roster used for that reconstruction.

The terminal-versus-convergence comparison and selector composition analysis were not confirmatory hypotheses. They are labeled post-hoc throughout. They may suggest a selector-stopping mismatch, but they cannot change the H1-H4

decisions.

3.6 Persona-v2 dual-condition protocol

Persona v2 is frozen separately from the programmatic study. Its independent cluster is `persona_id`, represented in prose as 50 persona clusters. Each cluster contains a paired deficit row and control row. Provider, prompt condition, deficit-control arm, and item are repeated observations within the cluster; the analysis therefore treats provider as a repeated measure alongside the other within-cluster factors. Calls, retries, individual item answers, and model outputs are not independent sample units.

The grid crosses 25 failure anchors with low and high response-noise settings. The controlled manipulation condition exposes a general observable error policy and uses four calibration items, but its primary outcomes are the four-state answer composition (correct, incorrect, abstention, and technical/schema failure), conditional accuracy among complete non-abstaining answers, and paired deficit-control shifts. The blind condition removes target misconception labels and target options, reuses the calibration items, and adds up to 21 family-distinct diagnostic items. It reports terminal-answer agreement, technical/schema failure, abstention, and two distinct repeat measures: answer-category agreement and canonical complete-output equality. The latter requires equality of the complete normalized response, not only the selected answer category. Every response uses a text-only modality.

Before provider observation, exact item/failure/option consensus was achieved for only 6 of 100 calibration rows. Because that coverage was below the frozen minimum, target-misconception hit rate was removed from the confirmatory Persona-v2 analysis. General target-policy compliance is therefore not a confirmatory outcome. The six mapped rows may appear only as a sparse descriptive target-option check with an explicit denominator and support hash. This degradation is an input fact, not a provider outcome.

The smoke pilot uses two providers and five clusters, is stored separately, and is excluded from every main estimate. Main collection attempts six frozen providers. Intervals use a 10,000-resample persona-cluster bootstrap that preserves all repeated provider, prompt-condition, deficit-control, and item observations for a sampled cluster. The pre-specified cross-model LLM adjudication protocol would receive blind public prompts and candidate outputs without target labels, correct options, or mapping status. Allowed labels include unknown and insufficient evidence. Actual judge completion is bound in Results; unavailable judges produce no labels or pairwise statistic. Any agreement and disagreement examples are descriptive, and no realism or authenticity score is formed.

Exact temporal, model, request, and runtime provenance appears in Results.

3.7 Illustrative prescription analysis

The parent validated library contains 68 trusted node-chunk assignments spanning 59 unique chunks across 13 nodes. Its exact overlap with the H1-H4 target catalog is much smaller: two fixed target strata, eight eligible chunks, and three distinct physical video sources. This overlap, not the parent-library count, defines the P2 analytic support.

P2 used matched-generator, budget-15 posteriors for Arms A, B, and C and one-hot truth vectors for the oracle. Within each arm, 16 equally weighted truth pairs crossed the 50 replicate margins from each target, giving 40,000 product-form integration terms per arm and 160,000 fixed profile rows overall. Each truth cell has weight 1/16 and each within-cell cross-product term has weight 1/2,500. These terms are not independent observations, joint learners, or an empirical prevalence distribution.

Under one 600-second budget, the greedy selector maximized posterior-weighted binary coverage of target-state role slots, with physical-source deduplication and saturation after the first compatible chunk. It ranked feasible choices by marginal utility per second, then marginal utility, shorter duration, and lexicographic chunk identifier. This is a problem-specific heuristic related to budgeted coverage (Samir Khuller et al. 1999); no approximation guarantee, optimality claim, or exact-solver comparison is made.

Reported fields separate mechanically mismatched selected minutes, missed available-supply minutes, unused budget, and diagnostic structural failure. For Arm C, one of the two target nodes is structurally invalid at budget 15; it is masked without reading or imputing its stored belief, and the bound report must retain the exact failed-node numerator and denominator (1/2). `unobtainable_supply_minutes` remains null, not zero, because the frozen model defines no role-compatible P/U dose. The finite-design means are illustrative; the 160,000 integration terms receive no learner sample size or population-inferential confidence interval. Any separately bound bootstrap interval describes simulator Monte Carlo variability only.

3.8 AI-assisted research workflow

Generative AI systems supported study-design critique, software implementation and testing, simulation execution, data analysis, figure generation, and adversarial manuscript review, and generated the initial full manuscript prose and subsequent revision proposals. AI-generated suggestions did not enter the confirmatory result surface by prose alone. H1-H4 claims were checked against frozen analysis rules, immutable hashes, machine-generated metric records, and targeted regression tests. Provider outputs in Persona v2 are research objects and are distinct from AI assistance used to conduct the work.

The named systems were OpenAI Codex (GPT-5.6 Sol) for implementation, testing, analysis scaffolding, and manuscript drafting/revision, and Anthropic Claude Code (Claude Opus 4.8) for project-direction critique, review, and revision proposals. These names describe the recorded tools used in this working version; the final submission must preserve the exact product/version strings required by the selected journal and the actual session records.

The human author set project direction and claim boundaries and retains responsibility for source review, result interpretation, journal-policy compliance, and final text. AI systems were not treated as authors, participants, or external reference standards. Before submission, the human author must critically review and revise the complete manuscript and the disclosure must name the systems, model versions, and uses in the formula required by the selected journal. No credential, private prompt, or machine-local location is part of the manuscript evidence surface.

4. Results

4.1 Execution integrity and analysis populations

32400 intended journeys; 30800 valid; 1600 structural failures; 0 schema-invalid. The full analysis support contains 27 targets; the mechanically eligible H1/H2 support contains 23 targets.

All results below are simulated finite-budget estimates. Matched and misspecified conditions remain separate. H1-H4 decisions are reproduced from the frozen decision rules rather than reassigned from the apparent favorability of a point estimate. Their intervals describe Monte Carlo uncertainty conditional on the fixed 27-target catalog or stated subset; they do not support target-population or learner-population inference.

Hypothesis	Frozen decision
H1	partially supported
H2	not supported
H3	supported
H4	direction-only check passed

4.2 Prerequisite-gap rescue and reasoning-chain harm

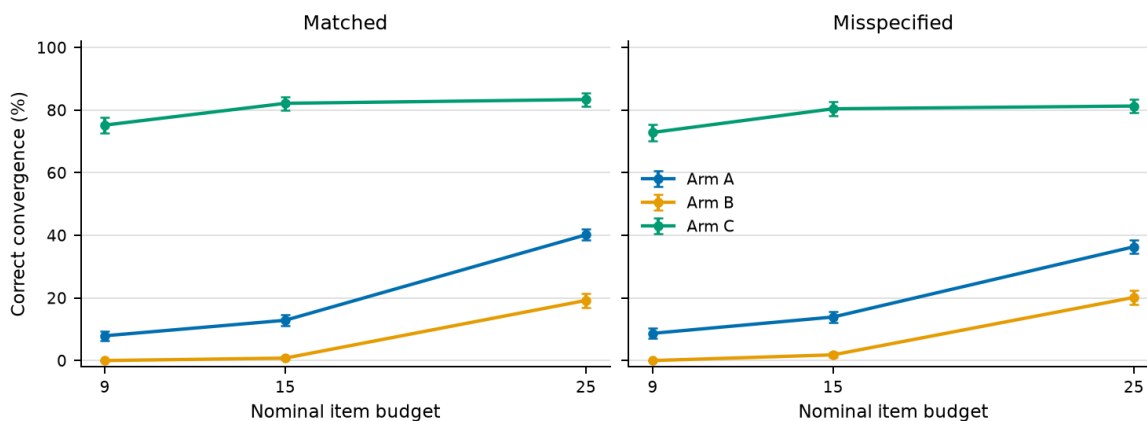
At budget 15 on the 23-target eligible stress support, Arm A P-state correct confident convergence was 148/1,150 (12.9%) (95% CI 11.2% to 14.5%), versus 9/1,150 (0.8%) (95% CI 0.3% to 1.3%) for Arm B. The paired A-minus-B contrast was 139/1,150 paired contrast; 12.1 percentage points; 95% CI 10.3% to 13.8%. The direction favored adaptive probing, but the Arm-A rate did not reach the frozen 50% threshold. H1 was partially supported.

On the no-repeat common support (4 targets; 200 paired cases), Arm A was 1/200 (0.5%) and Arm B was 0/200 (0.0%); the contrast was 1/200 paired contrast; 0.5 percentage points; 95% CI 0.0% to 1.5%. This jointly changed support-and-repeat estimand was much weaker, so its difference from the broad estimate cannot be attributed to repetition alone.

At budget 9 on the 23-target support, C-state misdiagnosis was 430/1,150 (37.4%) for Arm A, 318/1,150 (27.7%) for Arm B, and 495/1,150 (43.0%) for Arm C. C minus A was 65/1,150 paired contrast; 5.7 percentage points; 95% CI 2.3% to 9.0%. A minus B was 112/1,150 paired contrast; 9.7 percentage points; 95% CI 7.2% to 12.3%, exceeding the frozen +5-point no-harm margin. H2 was not supported.

On the 9-target no-repeat common support, the corresponding rates were 188/450 (41.8%) for A, 128/450 (28.4%) for B, and 197/450 (43.8%) for C. C minus A was 9/450 paired contrast; 2.0 percentage points; 95% CI -3.3% to 7.6%, whereas A minus B was 60/450 paired contrast; 13.3 percentage points; 95% CI 9.1% to 17.6%.

P-state rescue across item budgets



Points are target-equal estimates; error bars: 95% CI; n = 1,150 journeys per arm-budget cell.

Figure 1. P-state correct convergence across item budgets. The figure is reused from the verified programmatic artifact and does not include Persona-v2 evidence.

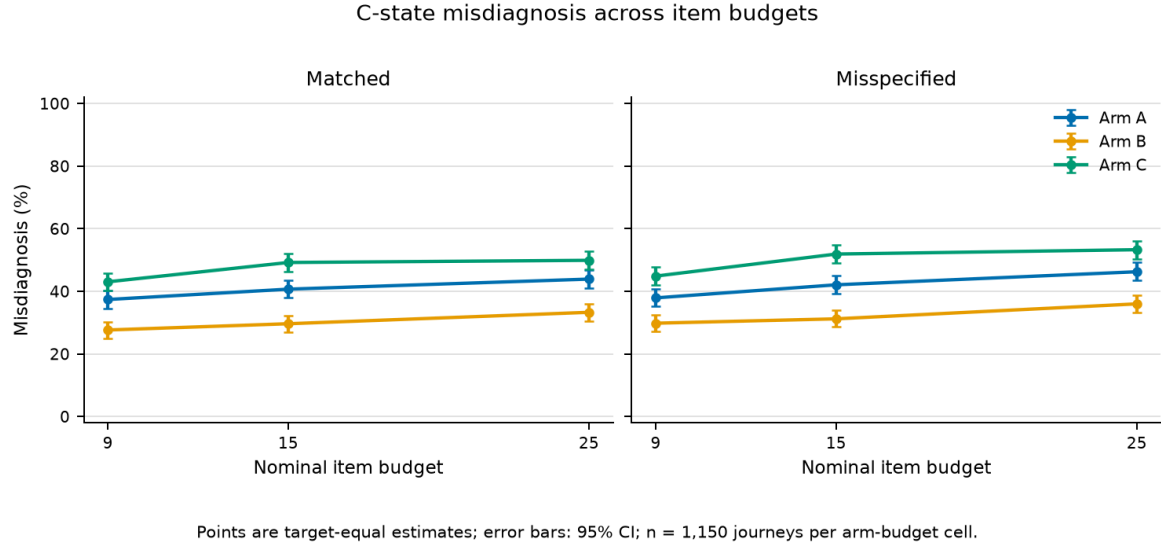


Figure 2. C-state misdiagnosis across item budgets. Arm labels refer only to the programmatic study.

4.3 Adaptive sanity check and misspecification

On the full 27-target matched set at budget 15, Arm A exceeded Arm B in terminal accuracy by 542/5,400 paired contrast; 10.0 percentage points; 95% CI 8.9% to 11.2%. Median time to confidence was 8 items for A (95% CI 8 to 8) and 11 items for B (95% CI 10 to 11). H3 was supported; this is an implementation sanity check, not an algorithmic novelty claim.

Under the misspecified generator, P-state A minus B was 139/1,150 paired contrast; 12.1 percentage points; 95% CI 10.2% to 14.0%; C-state C minus A was 80/1,150 paired contrast; 7.0 percentage points; 95% CI 3.7% to 10.3%; and C-state A minus B was 93/1,150 paired contrast; 8.1 percentage points; 95% CI 5.5% to 10.7%. Matched minus misspecified degradation was 0.0 points for H1 (95% CI -2.6% to 2.5%), -1.3 points for fixed-quota harm (95% CI -6.0% to 3.5%), and 1.7 points for adaptive minus local (95% CI -2.1% to 5.3%). H4 was supported under its direction-only rule; this does not establish robustness to untested generator families.

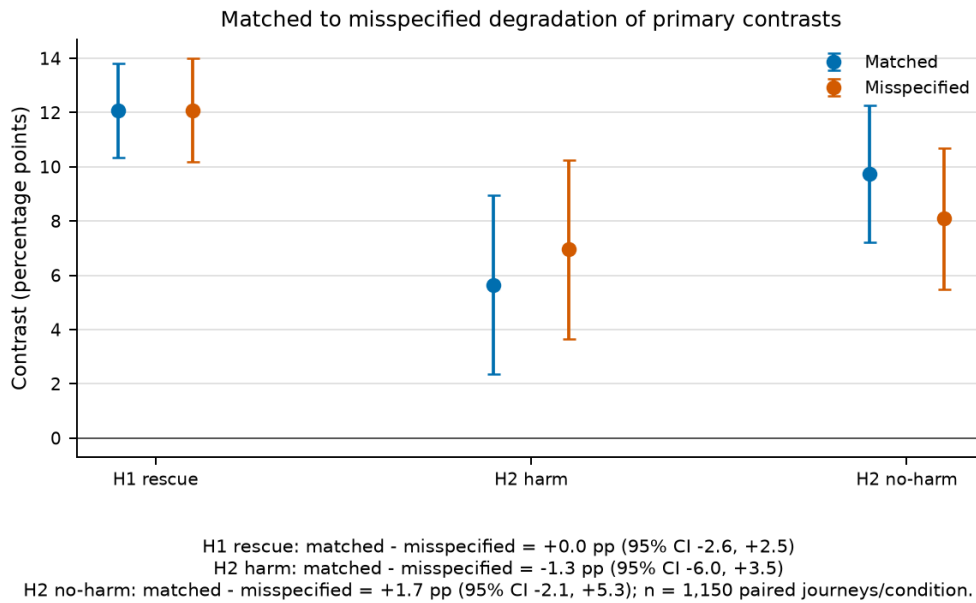


Figure 3. Matched and misspecified contrast estimates. The perturbation is one declared synthetic sensitivity condition.

4.4 Same-support terminal and convergence estimates

Terminal argmax accuracy and correct confident convergence answered different questions. The former asked whether the correct state had the largest posterior at the end of the budget. The latter additionally required the production confidence and posterior-gap thresholds. Table 1 preserves support identity for both views.

Pair ID	Support	Terminal accuracy	Correct convergence	Difference
full_27_terminal_vs_correct_convergence	full	1,131/1,350 (83.8%)	168/1,350 (12.4%)	71.3 pp
eligible_23_terminal_vs_correct_convergence	eligible	1,043/1,150 (90.7%)	148/1,150 (12.9%)	77.8 pp

Rows are not cross-compared: target-set hashes are full_27_terminal_vs_correct_convergence=726c098269012f371cc36c25691c1dfd4ad6963e2f0bba14a32c888d5fdbca9e; eligible_23_terminal_vs_correct_convergence=1220fc5d686df0d5b2d866c4587dca5423fafa60d73f1f8465f6ed703e85c016. The full-support difference is arithmetic only because no dedicated bootstrap interval is bound. The eligible-support difference retains its paired interval. Post-hoc selector composition on the eligible support averaged 1.43 direct and 11.95 prerequisite items by budget 15; prerequisite share was 83.8% (95% CI 82.6% to 84.9%). This selector-stopping mismatch is exploratory and does not revise H1.

The selector composition is interpreted only as a post-hoc mechanism clue. We refer to the observed pattern as a **selector-stopping mismatch**. It is exploratory, not a randomized mechanism test, and it does not revise H1.

4.5 Persona-v2 dual-condition results

Formal Persona-v2 main results use 50 persona_id clusters, provider as a repeated measurement, text-only modality, and disjoint pilot/main task rosters. These simulated response-channel estimates measure instruction following and robustness, not human behavioral validity. 3/6 providers were eligible for both controlled and blind aggregates.

Formal collection provenance. Phase first observation: 2026-07-15T20:27:09.136311Z UTC; provider-event ledgers span 2026-07-15T20:27:10.561439Z to 2026-07-15T23:20:17.307143Z UTC (execution-evidence, not per-response, timestamps). All requests set temperature=0; no explicit top_p or seed. Requested models (returned IDs matched where supplied): deepseek=deepseek-v4-pro; glm=glm-4-plus; kimi=moonshot-v1-128k; minimax=abab6.5s-chat; doubao=doubao-seed-2-0-mini-260428; tongyi=qwen-max. Runtime groups (tokens initial/retry [observed]; timeout s/worker cap/attempts; breaker; backoff s/cooldown s; jitter %): deepseek=1,024/2,048 [1,024+2,048]; 90/4/4; 3; 1-30/120; 25. doubao=1,024/2,048 [1,024]; 120/2/4; 3; 2-60/180; 25. glm/kimi/minimax/tongyi=512/1,024 [512]; 60/4/3; 3; 1-30/120; 25.

Provider lifecycle	Records	Controlled status	Blind status	Exclusion reasons
deepseek	4/1,528	controlled excluded	blind excluded	minimum_complete_clusters, repeated_failure
glm	4/1,528	controlled excluded	blind excluded	minimum_complete_clusters, repeated_failure
kimi	1,528/1,528	controlled eligible	blind eligible	none
minimax	56/1,528	controlled excluded	blind excluded	minimum_complete_clusters, repeated_failure
doubao	1,528/1,528	controlled eligible	blind eligible	none
tongyi	1,528/1,528	controlled eligible	blind eligible	none

Controlled outcome	paired	Oriented effect	95% cluster CI	Denominator
Conditional accuracy	answer	13.4% (control_minus_deficit)	9.8% to 17.4%	paired persona denominator 50-50
Correct-response yield		14.2% (control_minus_deficit)	10.7% to 18.0%	paired persona denominator 50-50
Incorrect-response yield		13.2% (deficit_minus_control)	9.5% to 17.2%	paired persona denominator 50-50
Abstention yield		1.0% (deficit_minus_control)	0.3% to 1.8%	paired persona denominator 50-50
Technical/schema-failure yield		0.0% (deficit_minus_control)	0.0% to 0.0%	paired persona denominator 50-50

Blind/exploratory outcome	Estimate	Denominator/interval
Blind technical/schema failure rate	0.0%	0.0% to 0.0%
Blind terminal exact agreement	48.0% to 79.0%	3 provider pairs; 100 paired terminal subjects each
Repeat answer-category stability	95.0%	87.0% to 100.0%
Repeat canonical complete-output stability	11.7%	3.7% to 20.4%
Automated error coding (exploratory)	Unavailable	Automated error coding was unavailable: GPT judge failed and Claude judge was unavailable; pairwise judge agreement was not estimable

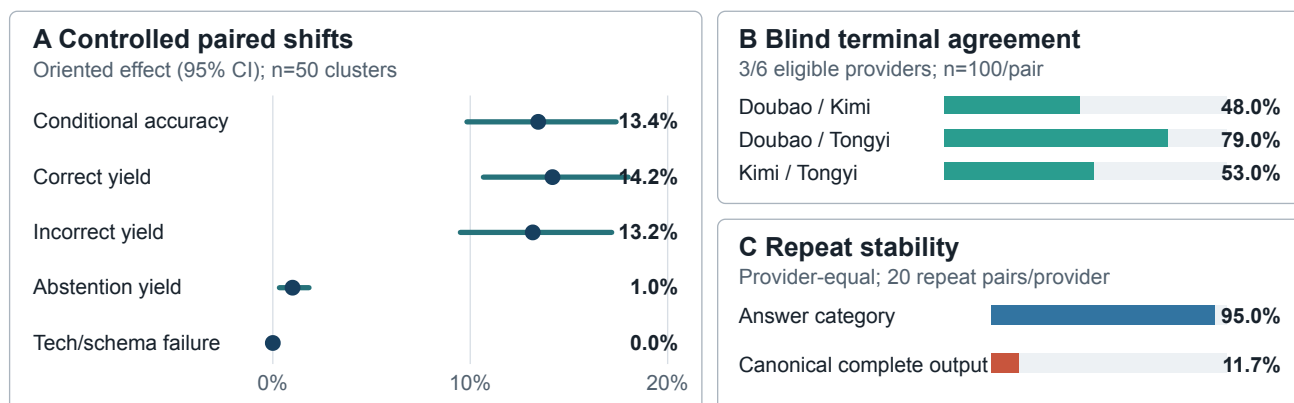


Figure. Compact Persona-v2 composite: controlled paired shifts, blind terminal agreement, and answer-category versus canonical complete-output repeat stability. Detailed provider panels remain in the supplement.

The table and compact figure report paired controlled answer-state composition and shifts, blind agreement, technical/schema failure, abstention, and repeat stability only for lifecycle-eligible provider cells. General target-policy compliance is not a confirmatory outcome; the six mapped rows remain sparse descriptive evidence. These are response-channel stress estimates, not evidence of human behavioral validity.

4.6 Illustrative prescription results

Illustrative P2 is supply-bound to 13 nodes and 68 trusted exact node-chunk assignments spanning 59 unique chunks. The fixed analytic overlap contains 2 targets, 8 candidate rows, and 3 physical sources under a 600-second analytic budget. This illustrative analysis does not estimate learning benefit or external remediation quality.

The specification-bound legacy output is retrospectively byte-anchored; its historical computation order is not cryptographically proven.

P/U role-compatible dose minutes are unavailable and remain null because the frozen library has no role-compatible dose definition for those states.

Arm	Mechanically mismatched selected min	Missed available-supply min	Diagnostic structural-failure node fraction
oracle	0.000	0.000	0.000
A	3.298	0.824	0.000
B	4.030	0.756	0.000
C	1.525	2.067	0.500

Arm C's failed-node fraction of 0.500 is a diagnostic structural failure and is retained intention-to-treat; it is not a zero-cost prescription. Missed available-supply minutes instead quantify supply scarcity within the fixed trusted library.

The parent boundary is 13 nodes, 68 trusted node-chunk assignments, and 59 unique chunks, but the analytic support is fixed at two target strata, eight eligible chunks, and three physical sources. The bound report must preserve the 160,000 weighted product-form integration rows as non-independent analytic terms, retain Arm C's one structurally failed target out of two, and leave `unobtainable_supply_minutes` null. These fields do not measure learning benefit, harm, or learner-population effects.

5. Discussion

5.1 Main interpretation

The primary experiment produced a mixed but informative result. Belief-triggered probing improved correct P convergence relative to a local-only ladder on the broad eligible stress support, yet the absolute convergence rate remained far below the

frozen criterion and weakened further on the small no-repeat common support. This is not a successful identification result masked by a strict criterion. It is evidence that the tested response channels and stopping thresholds did not reliably turn a directional advantage into an operationally confident diagnosis.

The C-state result is equally important. Fixed prerequisite insertion was worse than the adaptive policy on the primary support, matching the motivating direction. But the adaptive policy was itself materially worse than the local-only ladder, and the pre-specified no-harm condition failed. A narrative focused only on C minus A would therefore be incomplete. The three-arm design shows that avoiding a rigid quota did not imply that the adaptive policy was harmless.

H3 confirmed that the adaptive path improved aggregate terminal accuracy and reached confidence sooner than the local ladder across the full state grid. That advantage coexisted with poor P correct convergence. Aggregate adaptive efficiency can therefore hide a state-specific failure that matters for remediation routing. The direction-only H4 check found that the predicted rescue and fixed-quota harm directions survived one generator perturbation, but the design provides no warrant to generalize beyond that family.

5.2 Why terminal accuracy is insufficient

The largest descriptive finding was not an arm contrast but the same-support gap between terminal state ranking and confident convergence. On both the full and eligible supports, Arm A usually placed P first by the end of budget 15. It rarely satisfied the confidence-based stop. A product surface that displays only the argmax could therefore present a decisive-looking label where the production rule itself would continue gathering evidence or abstain.

The post-hoc selector composition makes this gap plausible. EIG repeatedly chose the prerequisite response channel, so the issue was not simply that the selector ignored prerequisite evidence. Selection optimized expected posterior information one item at a time; stopping required an absolute confidence and separation condition. Those objectives need not agree under sparse, overlapping likelihoods. The result motivates a direct audit of selector and stopper compatibility, not a claim that EIG is generally defective.

Operationally, three outputs should remain distinct: the terminal argmax, whether the confidence rule passed, and the action taken when it did not. A remediation system can use abstention, ask a different response type, or route to a low-risk general review when confidence remains insufficient. The present study does not establish which fallback improves outcomes, but it identifies the cases where a fallback is needed under the model.

5.3 Implications for response-channel stress testing

Programmatic simulation supplies known generating states and exact pairing, which makes it the strongest evidence in this paper for internal behavior. It cannot answer whether language models or people produce the assumed response patterns. Persona v2 addresses only the narrower question of response-channel robustness: do controlled prompts produce paired deficit-control shifts in answer-state composition, and do blind outputs remain stable when target labels are hidden?

The two-condition separation prevents prompted answer-state shifts from being mistaken for construct validity. Even strong shifts would not show that the induced errors resemble those of students. Blind cross-provider agreement may reveal a stable model response pattern; it would not establish that the pattern is educationally correct. Clustering by persona rather than by call prevents retries and repeated provider observations from inflating the sample size.

The pre-observation mapping degradation also changes what can be claimed. With exact consensus for only six calibration rows, a target-specific misconception hit rate would be dominated by ambiguous mappings. Removing it from the confirmatory family before provider observation is more informative than forcing a noisy metric into the paper. It leaves correctness, abstention, schema validity, terminal agreement, and answer-category and canonical complete-output repeat stability as separate response-channel outcomes. Only lifecycle-eligible providers contribute to these aggregates; their exact roster and denominators are reported rather than generalized to all attempted providers.

5.4 From posterior error to prescription constraints

The downstream prescription illustration expresses diagnostic error in a finite time currency: mechanically mismatched selected minutes and missed available-supply minutes. This translation is model-defined. It does not show that watching a selected chunk changes knowledge or that unmodeled P/U demand equals zero.

The parent library spans 13 nodes, 68 trusted node-chunk assignments, and 59 unique chunks, but the exact analytic overlap contains only two target strata, eight chunks, and three physical sources. Arm C's inability to open one of those two nodes is a structural failure, not a zero-cost outcome. The illustration therefore retains structural failure, missed available supply, unused budget, and the null unobtainable-dose field separately. Its weighted product-form terms are integration points, not learners or prevalence observations.

5.5 Claim boundaries and use

The results support an internal audit conclusion: selection efficiency, terminal accuracy, and confidence-based convergence are separable, and a policy that improves one state contrast can still violate a no-harm condition for another state. They do not support a claim that a learner was correctly diagnosed, that a remediation was pedagogically appropriate, or that the system improved achievement.

The intended use of the evidence is engineering triage. It identifies where richer response channels, better item-family coverage, altered stopping logic, or explicit abstention deserve evaluation before a larger intervention study. It also supplies a locally machine-reproducible negative result conditional on the fixed catalog: under the tested finite budgets, the P state remained poorly converged despite an adaptive rescue contrast. This is the precise sense in which the study documents budget-limited confident-convergence failure.

6. Limitations

1. **Simulation-only evidence.** No human participants contributed outcome data. The known generating state makes internal error analysis possible but cannot establish how people respond, learn, or interpret the diagnostic interface.
2. **Model circularity.** The matched generator uses the production likelihoods that inference assumes. The misspecified generator breaks that identity only through one frozen slip, guess, and ability-offset family, leaving many plausible response processes untested.
3. **Static diagnostic state.** The generating state is fixed within a journey. The experiment does not model knowledge acquisition, forgetting, transfer, or retention, and its results should not be read as longitudinal evidence.
4. **Text-only and multiple-choice-heavy channels.** The programmatic catalog is dominated by binary scored traces, and Persona v2 is explicitly text-only. The study does not test diagrams, experimental apparatus, open derivations, or the long-chain reasoning formats that motivated part of the instrument audit.
5. **Narrow curricular coverage.** Only 27 of 135 graph nodes enter the full programmatic grid, and only 23 satisfy the prerequisite-family gate for H1/H2. Important chemistry domains and difficult integrated problems remain outside the evidence base. Targets were fixed rather than sampled, so bootstrap intervals quantify simulator Monte Carlo uncertainty conditional on this catalog; they are not target-population, learner-population, or school-population intervals.
6. **Sparse pools and structural failures.** Family replenishment permits repeats in the broad stress estimand, while no-repeat common supports are small. Arm C also fails structurally for four targets, so comparisons are not uniformly available across the catalog.
7. **Stopping-rule specificity.** The terminal-versus-convergence gap depends on the current posterior confidence and separation thresholds. Other calibrated stopping rules may behave differently, and the post-hoc explanation has not been tested by a frozen stopping-rule intervention.
8. **No human reference labels for response coding.** The Persona-v2 protocol specified blind automated adjudication with unknown and insufficient-evidence options. The bound Results section reports whether any judge completed; no pairwise statistic is formed without two distinct model families, and unavailable coding remains unavailable. Automated coding cannot establish semantic correctness or behavioral realism.
9. **Independent item responses.** Persona-v2 elicits each response through a frozen response-channel protocol rather than modeling persistent memory or learning across a session. The independent unit is 50 persona clusters, with provider, prompt condition, deficit-control arm, and item repeated within cluster; provider calls are not additional units. Provider models and APIs can also drift over time, limiting temporal reproducibility despite recorded model identities.
10. **Constrained prescription supply.** The illustrative remediation analysis is drawn from a 13-node library with 68 trusted node-chunk assignments spanning 59 unique chunks, but its exact support is only two target strata, eight chunks, and three physical sources. Product-form integration terms are not learners, and mechanically mismatched selected or missed available-supply minutes describe only this fixed utility model.

7. Conclusion

This study audited a chemistry diagnostic system at the point where answer evidence becomes a model-defined state decision. Belief-triggered prerequisite probing produced a positive P-state rescue contrast relative to a local-only ladder, but it did not reach the frozen absolute convergence criterion. Fixed prerequisite insertion harmed C relative to the adaptive arm, while the adaptive arm also failed its no-harm test against the local ladder. Aggregate adaptive accuracy and speed improved, and the two directions required by the pre-specified direction-only H4 check persisted under one declared misspecification family.

Most importantly, correct terminal state ranking and correct confident convergence were far apart on both legitimate supports. That gap, together with the post-hoc selector composition, motivates treating item selection and stopping as separate objects of evaluation. In the audited implementation, the terminal argmax should not be converted into a precise remediation label without also reporting whether the separate confidence rule passed.

Persona-v2 response-channel estimates and the P2 prescription illustration extend the audit with secondary robustness and supply-bound downstream-cost evidence. They do not revise the frozen H1-H4 decisions, establish human behavioral validity, or turn this internal simulation audit into a study of human learning.

Declarations

Ethics and participant involvement

No human participants were enrolled and no personally identifying information was collected for the experiments reported here. Programmatic responses and language-model outputs are labeled as simulated evidence. Any future study involving students or human raters requires a separate ethics, consent, and data-governance determination.

Data and code availability

The local reproducibility package contains frozen analysis plans, versioned source, hash-bound input manifests, deterministic seeds, raw simulated response envelopes, metric registries, figure artifacts, and regression tests. Public release remains gated on a repository destination and a rights review covering question text, video metadata, and derived artifacts. Any submission statement must name the approved repository or access procedure. Credentials, private provider payloads, and material without distribution rights are excluded.

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Competing interests

The author declares no competing financial interest. The audited system was developed within the same project, which creates an investigator-interest risk addressed through pre-observation freezing, negative-result retention, and machine-bound reporting.

Declaration of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used for study-design critique, software implementation and testing, simulation execution, data analysis, figure generation, drafting and revising prose, and adversarial manuscript review. They generated the initial full manuscript prose and subsequent revision proposals. Outputs were checked against frozen contracts, source artifacts, automated tests, and stated claim boundaries. The systems were OpenAI Codex (GPT-5.6 Sol) and Anthropic Claude Code (Claude Opus 4.8). AI systems were not treated as authors, human participants, or external reference standards. Before submission, the human author must critically review and revise the complete manuscript and assume responsibility for its accuracy, integrity, interpretation, policy compliance, and final wording. The final disclosure must name the systems and model versions and follow the selected journal's current formula without implying that one publisher policy applies universally.

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